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PROTECTION CIRCUIT MODULE AND BATTERY MODULE COMPRISING THE SAME

Abstract

Disclosed are a protection circuit module and a battery module capable of detecting a trigger cell in which heat propagation occurs within the battery module, even when heat propagation occurs in a battery cell that does not have a temperature sensor. The protection circuit module includes: a substrate; a wire disposed on the substrate to form an opening in at least a portion of the wire and the wire including a first metal; a solder including a second metal having a melting point lower than a melting point of the first metal and the second metal being inserted into at least a portion of the opening to join end of the wire separated by the opening; and a cover forming at least one hole and covering at least a portion of the solder.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This present application claims priority to and the benefit under 35 U.S.C. § 119(a)-(d) of Korean Patent Application No. 10-2024-0032580, filed on Mar. 7, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference.

FIELD

[0002] The technology described herein relates to a protection circuit module and/or a battery module including the same.

BACKGROUND

[0003] A secondary battery can be charged and discharged unlike a primary cell that may not be repeatedly charged. Low-capacity secondary batteries are typically used in small, portable electronic devices, such as smartphones, feature phones, notebook computers, digital cameras, and camcorders, and high-capacity secondary batteries are widely used as power sources for motors in hybrid and electric vehicles and as power storage cells. Such a secondary battery includes an electrode assembly, a case receiving the electrode assembly, and electrode terminals connected to the electrode assembly.

[0004] This section is intended only to provide a better understanding of the background of the technology described herein and thus may include information which is not necessarily prior art.

SUMMARY

[0005] It is one aspect of the technology described herein to provide a protection circuit module and/or a battery module capable of detecting a trigger cell (cell from which heat propagates) having an increased temperature upon increase in temperature of each of a plurality of battery cells.

[0006] It is another aspect of the technology described herein to provide a protection circuit module and/or a battery module capable of detecting a trigger cell without applying a temperature sensor.

[0007] For example, the technology described herein enables detection of a trigger cell among battery cells without applying temperature sensors to all of the battery cells.

[0008] It is a further aspect of the technology described herein to provide a protection circuit module and/or a battery module capable of detecting a trigger cell while reducing manufacturing costs.

[0009] The above and other aspects and features of the technology described herein will become apparent from the following description of embodiments of the technology described herein.

[0010] In accordance with one aspect of the technology described herein, a protection circuit module includes: a substrate; a wire disposed on the substrate to form an opening in at least a portion of the wire and the wire including a first metal; a solder including a second metal having a melting point lower than a melting point of the first metal and the second metal being inserted into at least a portion of the opening to join ends of the wire separated by the opening; and a cover forming at least one hole and covering at least a portion of the solder.

[0011] In accordance with another aspect of the technology described herein, a battery module includes: a plurality of battery cells; a housing for receiving the plurality of battery cells; and a protection circuit module disposed at one side of the plurality of battery cells, the protection circuit module including: a substrate; a wire disposed on the substrate to form an opening in at least a portion of the wire and the wire including a first metal; a solder including a second metal having a melting point lower than a melting point of the first metal and the second metal inserted into at least a portion of the opening to join ends of the wire separated by the opening; and a cover

forming at least one hole and covering at least a portion of the solder.

[0012] In some embodiments, the at least one hole is placed on the solder and provides a migration path for vaporized second metal upon vaporization of the second metal.

[0013] In some embodiments, the opening has a width of 1 mm to 4 mm.

[0014] In some embodiments, the melting point of the second metal is 200° C. or higher.

[0015] In some embodiments, the second metal comprises at least one material selected from the group consisting of Al, Mg, Sb, Zn, Pb, Cd, Bi, and Sn.

[0016] In some embodiments, the cover comprises at least one material selected from the group consisting of polyethylene naphthalate (PEN), polyimide (PI), polyethylene terephthalate (PET), polytetrafluoroethylene (PTFE), polycarbonate (PC), and polyethersulphone (PES).

[0017] In some embodiments, the cover has a thickness of 0.015 mm to 0.060 mm.

[0018] In some embodiments, an area ratio of the solder to the at least one hole to the solder ranges from 0.4 to 0.65 in top plan view.

[0019] In some embodiments, the protection circuit module further comprises a connection plate electrically connected to a battery cell, wherein the opening is formed within 40 mm from the connection plate.

[0020] According to the technology described herein, it is possible to provide a protection circuit module and/or a battery module capable of detecting a trigger cell from which heat propagates even when heat propagates from a battery cell that is not provided with a temperature sensor.

[0021] According to the technology described herein, it is possible to provide a protection circuit module and/or a battery module capable of blocking external overcurrent in the event of heat propagation from a trigger cell.

[0022] According to the technology described herein, it is possible to provide a protection circuit module and/or a battery module capable of securing safety performance while reducing manufacturing costs.

[0023] However, aspects and features of the technology described herein are not limited to those described above and other aspects and features not mentioned will be clearly understood by those skilled in the art from the detailed description given below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The following drawings attached to this specification illustrate embodiments of the technology described herein and further describe aspects and features of the technology together with the detailed description of the technology. Thus, the technology described herein should not be construed as being limited to the drawings:

[0025] FIGS. **1** to **4** are schematic views of a lithium battery, according to one embodiment of the technology described herein;

[0026] FIG. **5** is a perspective view of a battery module according to one embodiment of the technology described herein;

[0027] FIG. **6** is a schematic cross-sectional view of a fuse according to one embodiment of the technology described herein;

[0028] FIG. **7** is a graph depicting temperature change of a single battery cell and other battery cells adjacent thereto over time upon increase in temperature of the corresponding battery cell;

[0029] FIG. **8** is a schematic cross-sectional view of a fuse according to one embodiment of the technology described herein;

[0030] FIG. **9** is a schematic top plan view of the fuse according to the embodiment of the technology described herein;

[0031] FIG. **10** is a top plan view of a battery module according to one embodiment of the

technology described herein; and
[0032] FIG. 11 is an enlarged view of Region A in FIG. 10.

DETAILED DESCRIPTION

[0033] Hereinafter, exemplary embodiments of the technology described herein will be described in detail with reference to the accompanying drawings. However, it should be understood that the following embodiments are provided by way of illustration and the technology described herein is not limited thereto and is defined only by the appended claims and equivalents thereto.

[0034] When an arbitrary element is referred to as being disposed (or placed or placed) “above” (or “below”) or “on” (or “under”) a component, it may mean that the arbitrary element is placed in contact with the upper (or lower) surface of the component and may also mean that another component may be interposed between the component and any arbitrary element disposed (or placed or placed) on (or under) the component.

[0035] Throughout the specification, unless specified otherwise, each element may be singular or plural. In addition, throughout the specification, when “A and/or B” is stated, it means A, B or A and B, unless specified otherwise.

[0036] As used herein, “combinations thereof” may refer to mixtures, stacks, composites, copolymers, alloys, blends, and reaction products of components.

[0037] Unless otherwise defined herein, particle diameter may refer to an average particle diameter. In addition, the particle diameter means an average particle diameter (D50) that refers to a particle diameter corresponding to 50% by volume in a volume cumulative variation of corresponding particles. The average particle diameter may be measured by any method well known in the art, for example, by a particle size analyzer, a transmission electron microscope image, or a scanning electron microscope image. Alternatively, the average particle diameter (D50) may be measured by counting the number of particles in each particle diameter range using a device employing a dynamic light-scattering method to analyze data, followed by calculating the average particle diameter (D50) based on the analyzed data. Alternatively, the average particle diameter (D50) may be measured by laser diffraction. More specifically, in measurement by laser diffraction, target particles are dispersed in a dispersant, introduced into a commercially available laser diffraction particle analyzer (for example, Microtrac MT 3000), and irradiated with ultrasound waves of about 28 kHz at a power of 60 W, followed by calculating the average particle size (D50) corresponding to 50% by volume in the cumulative volume variation of the particles in the measurement device.

[0038] With development of technology, higher capacity batteries are required. Accordingly, a plurality of batteries may be electrically connected and used. For example, the batteries may be applied to electronic devices in the form of a battery module including a plurality of batteries and/or a battery pack including a plurality of battery modules. Here, the electronic devices refer to electronic devices requiring high power and/or high capacity, for example, electric vehicles.

[0039] A battery module includes a plurality of battery cells. Accordingly, a protection circuit module for managing the plurality of battery cells is required. The protection circuit module may check, for example, a charge state of each of the battery cells. Alternatively, the protection circuit module may measure, for example, a temperature of each of the battery cells. Alternatively, the protection circuit module may manage the plurality of battery cells by detecting, for example, whether the battery cells undergo overcurrent and/or short circuit.

[0040] The protection circuit module includes a fuse to perform all or some of the functions described above. The fuse measures the temperature of the battery cells, for example, through a temperature sensor, and interrupts power to each of the battery cells based on the measured temperature. However, in order to reduce manufacturing costs, a single temperature sensor is typically disposed per plurality of battery cells. Accordingly, when heat propagates from a certain battery cell not provided with the temperature sensor, there is difficulty detecting heat propagation from the battery cell.

[0041] FIG. 1 to FIG. 4 are schematic views of lithium secondary batteries according to

embodiments of the technology described herein.

Lithium Secondary Battery 100

[0042] Lithium secondary batteries may be classified into a cylindrical secondary battery, a faceted secondary battery, a pouch type secondary battery, a coin type secondary battery, and the like based on the shapes thereof. FIG. 1 to FIG. 4 are schematic views of lithium secondary batteries according to embodiments of the technology described herein, in which FIG. 1 shows a cylindrical secondary battery, FIG. 2 shows a faceted secondary battery, and FIG. 3 and FIG. 4 show pouch-type secondary batteries. Referring to FIG. 1 to FIG. 4, a lithium secondary battery 100 may include an electrode assembly 40 in which a separator 30 is interposed between a cathode 10 and an anode 20, and a case 50 that receives the electrode assembly 40 therein. The cathode 10, the anode 20, and the separator 30 may be immersed in an electrolyte (not shown). The lithium secondary battery 100 may include a sealing member 60 that seals the case 50, as shown in FIG. 1. In addition, as shown in FIG. 2, the lithium secondary battery 100 may include a cathode lead tab 11, a cathode terminal 12, an anode lead tab 21, and an anode terminal 22. Referring to FIG. 3 and FIG. 4, the lithium secondary battery 100 may include electrode tabs 70, that is, a cathode tab 71 and an anode tab 72, which act as electrical pathways conducting current formed in the electrode assembly 40 to the outside.

Cathode Material

[0043] As a cathode material, a compound allowing reversible intercalation and deintercalation of lithium ion (lithiated intercalation compound) may be used. Specifically, the cathode material may be at least one complex oxide of a metal selected from among cobalt, manganese, nickel and combinations thereof with lithium.

[0044] The composite oxide may be a lithium transition metal composite oxide.

[0045] Specifically, the lithium transition metal composite oxide may be a lithium nickel oxide, a lithium cobalt oxide, a lithium manganese oxide, a lithium iron phosphate compound, a cobalt-free nickel-manganese oxide, or a combination thereof.

[0046] By way of example, the composite oxide may be a compound represented by any of the following formulas: $\text{Li}_{a_1}\text{A}_{b_1}\text{X}_{c_1}\text{O}_{d_1}$ ($0.90 \leq a_1 \leq 1.8$, $0 \leq b_1 \leq 0.5$, $0 \leq c_1 \leq 0.05$); $\text{Li}_{a_2}\text{Mn}_{b_2}\text{X}_{c_2}\text{O}_{d_2}$ ($0.90 \leq a_2 \leq 1.8$, $0 \leq b_2 \leq 0.5$, $0 \leq c_2 \leq 0.05$); $\text{Li}_{a_3}\text{Ni}_{b_3}\text{X}_{c_3}\text{O}_{d_3}$ ($0.90 \leq a_3 \leq 1.8$, $0 \leq b_3 \leq 0.5$, $0 \leq c_3 \leq 0.5$, $0 < d_3 < 2$); $\text{Li}_{a_4}\text{Ni}_{b_4}\text{Mn}_{c_4}\text{X}_{d_4}\text{O}_{e_4}$ ($0.90 \leq a_4 \leq 1.8$, $0 \leq b_4 \leq 0.5$, $0 \leq c_4 \leq 0.5$, $0 < d_4 < 2$); $\text{Li}_{a_5}\text{Ni}_{b_5}\text{Co}_{c_5}\text{L}_{d_5}\text{G}_{e_5}\text{O}_{f_5}$ ($0.90 \leq a_5 \leq 1.8$, $0 \leq b_5 \leq 0.9$, $0 \leq c_5 \leq 0.5$, $0 \leq d_5 \leq 0.5$, $0 \leq e_5 \leq 0.1$); $\text{Li}_{a_6}\text{Ni}_{b_6}\text{G}_{c_6}\text{O}_{d_6}$ ($0.90 \leq a_6 \leq 1.8$, $0.001 \leq b_6 \leq 0.1$); $\text{Li}_{a_7}\text{Co}_{b_7}\text{G}_{c_7}\text{O}_{d_7}$ ($0.90 \leq a_7 \leq 1.8$, $0.001 \leq b_7 \leq 0.1$); $\text{Li}_{a_8}\text{Mn}_{b_8}\text{G}_{c_8}\text{O}_{d_8}$ ($0.90 \leq a_8 \leq 1.8$, $0.001 \leq b_8 \leq 0.1$); $\text{Li}_{a_9}\text{Mn}_{b_9}\text{G}_{c_9}\text{PO}_{d_9}$ ($0.90 \leq a_9 \leq 1.8$, $0 \leq b_9 \leq 0.5$); $\text{Li}_{a_{10}}(3-f)\text{Fe}_{b_{10}}(\text{PO})_{c_{10}}$ ($0 \leq f \leq 2$); and $\text{Li}_{a_{11}}\text{FePO}_{b_{11}}$ ($0.90 \leq a_{11} \leq 1.8$).

[0047] In the above formulas, A is Ni, Co, Mn, or a combination thereof; X is Al, Ni, Co, Mn, Cr, Fe, Mg, Sr, V, a rare-earth element, or a combination thereof; D is O, F, S, P, or a combination thereof; G is Al, Cr, Mn, Fe, Mg, La, Ce, Sr, V, or a combination thereof; and L^{sup.1} is Mn, Al, or a combination thereof.

[0048] In one embodiment, the cathode material may be a high nickel-content cathode material containing 80 mol % or more, 85 mol % or more, 90 mol % or more, 91 mol % or more, or 94 mol % to 99 mol % of nickel relative to 100 mol % of metal excluding lithium in the lithium transition metal complex oxide. The high nickel-content cathode material can achieve high capacity and thus can be applied to high capacity/high density lithium secondary batteries.

Cathode 10

[0049] The cathode 10 for the lithium secondary battery 100 may include a cathode current collector and a cathode material layer formed on the cathode current collector. The cathode material layer includes a cathode material and may further include a binder and/or a conductive material.

[0050] In one embodiment, the cathode may further include an additive capable of acting as a sacrificial cathode.

[0051] The cathode material may be present in an amount of 90 wt % to 99.5 wt % based on 100 wt % of the cathode material layer and each of the binder and the conductive material may be present in an amount of 0.5 wt % to 5 wt % based on 100 wt % of the cathode material layer.

[0052] The binder serves to attach cathode material particles to each other while attaching the cathode material to the cathode current collector. The binder may include, for example, polyvinyl alcohol, carboxymethylcellulose, hydroxypropylcellulose, diacetylcellulose, polyvinyl chloride, carboxylated polyvinyl chloride, polyvinyl fluoride, polymers including ethylene oxide, polyvinyl pyrrolidone, polyurethane, polytetrafluoroethylene, polyvinylidene fluoride, polyethylene, polypropylene, styrene-butadiene rubbers, (meth)acrylated styrene-butadiene rubbers, epoxy resins, (meth)acrylic resins, polyester resins, Nylon, and the like, without being limited thereto.

[0053] The conductive material serves to impart conductivity to the electrodes and may be any electrically conductive material that does not cause chemical change in cells under construction. The conductive material may include, for example, carbon materials, such as natural graphite, artificial graphite, carbon black, acetylene black, Ketjen black, carbon fiber, carbon nanofibers, carbon nanotubes, and the like; metal-based materials in the form of metal powders or metal fibers containing copper, nickel, aluminum, silver, and the like; conductive polymers, such as polyphenylene derivatives and the like; and mixtures thereof.

[0054] The cathode current collector may be Al foil, without being limited thereto.

Anode Material

[0055] The anode material includes a material allowing reversible intercalation/deintercalation of lithium ions, lithium metal, lithium metal alloy, a material capable of being doped to lithium and de-doped therefrom, or a transition metal oxide.

[0056] The material allowing reversible intercalation/deintercalation of lithium ions may include a carbon-based anode material, for example, crystalline carbon, amorphous carbon, or a combination thereof. The crystalline carbon may include, for example, graphite, such as natural graphite or artificial graphite, in amorphous, plate, flake, spherical, or fibrous form, and the amorphous carbon may include, for example, soft carbon, hard carbon, mesoporous pitch carbides, calcined coke, and the like.

[0057] The lithium metal alloy may be an alloy of lithium and a metal selected from among Na, K, Rb, Cs, Fr, Be, Mg, Ca, Sr, Si, Sb, Pb, In, Zn, Ba, Ra, Ge, Al and Sn may be used.

[0058] The material capable of being doped to lithium and de-doped therefrom may be an Si-based anode material or an Sn-based anode material. The Si-based anode material may be silicon, a silicon-carbon composite, $\text{SiO}_{0.5}$ ($0 < x \leq 2$), Si-Q alloys (where Q is selected from among alkali metals, alkali-earth metals, Group XIII elements, Group XIV elements (excluding Si), Group XV elements, Group XVI elements, transition metals, rare-earth elements, and combinations thereof), or combinations thereof. The Sn-based anode material may be Sn, SnO_x ($0 < x \leq 2$) (e.g., SnO_2), an Sn alloy, or a combination thereof.

[0059] The silicon-carbon composite may be a composite of silicon and amorphous carbon. According to one embodiment, the silicon-carbon composite may be prepared in the form of silicon particles having an amorphous carbon coating formed on the surface thereof. For example, the silicon-carbon composite may include secondary particles (cores) composed of primary silicon particles and an amorphous carbon coating layer (shell) formed on the surface of the secondary particle. The amorphous carbon may also be placed between the primary silicon particles such that, for example, the primary silicon particles are coated with amorphous carbon. The secondary particles may be dispersed in an amorphous carbon matrix.

[0060] The silicon-carbon composite may further include crystalline carbon. For example, the silicon-carbon composite may include a core containing crystalline carbon and silicon particles, and an amorphous carbon coating layer formed on the core.

[0061] The Si-based anode material or the Sn-based anode material may be used in combination with the carbon-based anode material.

Anode **20**

[0062] The anode **20** for the lithium secondary battery **100** may include an anode current collector and an anode material layer formed on the anode current collector. The anode material layer includes an anode material and may further include a binder and/or a conductive material.

[0063] For example, the anode material layer may include 90 wt % to 99 wt % of the anode material, 0.5 wt % to 5 wt % of the binder, and 0 wt % to 5 wt % of the conductive material.

[0064] The binder serves to attach the anode material particles to each other while attaching the anode material to the anode current collector. The binder may be a non-aqueous binder, an aqueous binder, a dry binder, or a combination thereof.

[0065] The non-aqueous binder includes polyvinyl chloride, carboxylated polyvinyl chloride, polyvinyl fluoride, ethylene propylene copolymers, polystyrene, polyurethane, polytetrafluoroethylene, polyvinylidene fluoride, polyethylene, polypropylene, polyamideimide, polyimide, or combinations thereof.

[0066] The aqueous binder may be selected from the group consisting of styrene-butadiene rubbers, (meth)acrylated styrene-butadiene rubbers, (meth)acrylonitrile-butadiene rubbers, (meth)acrylic rubbers, butyl rubbers, fluorinated rubbers, polyethylene oxide, polyvinyl pyrrolidone, polyepichlorohydrin, polyphosphazene, poly(meth)acrylonitrile, ethylene propylene diene copolymers, polyvinyl pyridine, chlorosulfonated polyethylene, latex, polyester resins, (meth)acryl resins, phenol resins, epoxy resins, polyvinyl alcohol, and combinations thereof.

[0067] When the aqueous binder is used as the anode binder, a cellulose-based compound capable of imparting viscosity may be further included. The cellulose-based compound may be a mixture of carboxymethylcellulose, hydroxypropyl methylcellulose, methylcellulose, or alkali metal salts thereof. The alkali metal may be Na, K, or Li.

[0068] The dry binder may be a fibrous polymeric material and may include, for example, polytetrafluoroethylene, polyvinylidene fluoride, a polyvinylidene fluoride-hexafluoropropylene copolymer, polyethylene oxide, or a combination thereof.

[0069] The conductive material serves to impart conductivity to the electrodes and may be any electronically conductive material that does not cause chemical change in cells under construction. Specifically, the conductive material may include, for example, carbon materials, such as natural graphite, artificial graphite, carbon black, acetylene black, Ketjen black, carbon fibers, carbon nanofibers, carbon nanotubes, and the like; metal-based materials in the form of metal powders or metal fibers containing copper, nickel, aluminum, silver, and the like; conductive polymers, such as polyphenylene derivatives and the like; or mixtures thereof.

[0070] The anode current collector may be selected from copper foil, nickel foil, stainless steel foil, titanium foil, nickel foam, copper foam, a conductive metal-coated polymer base, and combinations thereof.

Electrolyte (not Shown)

[0071] The electrolyte for the lithium secondary battery **100** includes a non-aqueous organic solvent and a lithium salt.

[0072] The non-aqueous organic solvent acts as a medium through which ions involved in electrochemical reaction of a cell can move.

[0073] The non-aqueous organic solvent may be a carbonate-based solvent, an ester-based solvent, an ether-based solvent, a ketone-based solvent, an alcohol-based solvent, a non-amphoteric solvent, or a combination thereof.

[0074] The carbonate-based solvents may include dimethyl carbonate (DMC), diethyl carbonate (DEC), dipropyl carbonate (DPC), methyl propyl carbonate (MPC), ethyl propyl carbonate (EPC), methyl ethyl carbonate (MEC), ethylene carbonate (EC), propylene carbonate (PC), and/or butylene carbonate (BC).

[0075] The ester-based solvents may include methyl acetate, ethyl acetate, n-propyl acetate, dimethyl acetate, methyl propionate, ethyl propionate, decanolide, mevalonolactone, valerolactone, caprolactone, and the like.

[0076] The ether-based solvents may include dibutyl ether, tetraglyme, diglyme, dimethoxyethane, 2-methyltetrahydrofuran, 2,5-dimethyltetrahydrofuran, tetrahydrofuran, and the like. In addition, the ketone-based solvent may include cyclohexanone and the like. The alcohol-based solvents may include ethyl alcohol, isopropyl alcohol, and the like, and non-amphoteric solvent may include nitriles, such as R—CN (where R is a straight, branched or cyclic hydrocarbon group having 2 to 20 carbon atoms and may include double bonds, aromatic rings, or ether groups); amides, such as dimethylformamide; dioxolanes, such as 1,3-dioxolane, 1,4-dioxolane, and the like; sulfolanes; and the like.

[0077] The non-aqueous organic solvent may be used alone or as a mixture thereof.

[0078] In use of the carbonate-based solvent, a mixture of a cyclic carbonate and a chained carbonate may be used, and the cyclic carbonate and the chained carbonate may be mixed in a volume ratio of 1:1 to 1:9.

[0079] The lithium salt is a substance that is soluble in an organic solvent and serves as a source of lithium ions in a battery, enabling operation of a basic lithium secondary battery while facilitating transfer of the lithium ions between the cathode and the anode. Examples of the lithium salts may include at least one selected from among LiPF_6 , LiBF_4 , LiSbF_6 , LiAsF_6 , LiClO_4 , LiAlO_2 , LiAlCl_4 , LiPO_2F_2 , LiCl , LiI , $\text{LiN}(\text{SO}_3\text{CF}_3)_2$, $\text{Li}(\text{FSO}_2)_2\text{N}$ (lithium bis(fluorosulfonyl)imide (LiFSI), $\text{LiC}_4\text{F}_9\text{SO}_3$, $\text{LiN}(\text{C}_x\text{F}_{2x+1}\text{SO}_2)(\text{C}_y\text{F}_{2y+1}\text{SO}_2)$ (where x and y are integers of 1 to 20), lithium trifluoromethane sulfonate, lithium tetrafluoroethane sulfonate, lithium difluorobis(oxalato)phosphate (LiDFOB), and lithium bis(oxalato) borate (LiBOB).

Separator 30

[0080] Depending on the type of lithium secondary battery 100, a separator 30 may be interposed between the cathode 10 and the anode 20. For such a separator 30, polyethylene, polypropylene, polyvinylidene fluoride, or at least two layers thereof may be used as well as mixed layers, such as a polyethylene/polypropylene bilayer separator, a polyethylene/polypropylene/polyethylene trilayer separator, a polyethylene/polyethylene/polypropylene trilayer separator, and the like.

[0081] The separator 30 may include a porous substrate and a coating layer that includes an organic material, an inorganic material, or a combination thereof on one or both surfaces of the porous substrate.

[0082] The porous substrate may be a polymer layer formed of a polymer selected from among polyolefins, such as polyethylene polypropylene, and the like, polyesters, such as polyethylene terephthalate, polybutylene terephthalate, and the like, polyacetal, polyamides, polyimides, polycarbonates, polyether ketones, polyarylether ketones, polyetherimides, polyamideimides, polybenzimidazole, polyethersulfone, polyphenylene oxides, cyclic olefin copolymers, polyphenylene sulfides, polyethylene naphthalate, glass fiber, Teflon, and polytetrafluoroethylene, copolymers thereof, or mixtures thereof.

[0083] The organic material may include a polyvinylidene fluoride polymer or a (meth)acrylic polymer.

[0084] The inorganic material may include inorganic particles selected from among Al_2O_3 , SiO_2 , TiO_2 , SnO_2 , CeO_2 , MgO , NiO , CaO , GaO , ZnO , ZrO_2 , Y_2O_3 , SrTiO_3 , BaTiO_3 , $\text{Mg}(\text{OH})_2$, boehmite, and combinations thereof, without being limited thereto.

[0085] The organic material and the inorganic material may be present in a mixed state in one coating layer or may be present in the form of a stack structure of a coating layer including the organic material and a coating layer including the inorganic material.

[0086] FIG. 5 is a perspective view of a battery module according to one embodiment of the technology described herein.

[0087] A battery module **1000** according to one embodiment of the technology described herein includes a plurality of battery cells **100**, a housing **1061**, **1062**, **1063**, **1064**, **1065** that receives the plurality of battery cells **100** therein, and a busbar electrically connecting at least some of the plurality of battery cells **100** to each other.

[0088] The plurality of battery cells **100** may include, for example, the battery cells illustrated in FIG. 1 to FIG. 4, and may be arranged in one direction and received within the housing **1061** to **1065**.

[0089] The housing **1061** to **1065** may include a pair of end plates **1061**, **1062** facing wide surfaces of the battery cells **100**, and side plates **1063** and a bottom plate **1064** each connecting the pair of end plates **1061**, **1062** to each other. The side plates **1063** may support side surfaces of each of the battery cells **100** and the bottom plate **1064** may support a bottom surface of each of the battery cells **100**. In addition, the pair of end plates **1061**, **1062**, the side plates **1063** and the bottom plates **1064** may be connected to one another by members, such as bolts **1065** or the like.

[0090] The battery module **100** includes terminals **1011**, **1012**, connection tabs **1020** connecting adjacent battery cells **100** to each other, and a protection circuit module **1030** connected at one side thereof to the connection tabs **1020**. The protection circuit module **1030** may be a battery management system (BMS). The connection tabs **1020** may be busbars.

[0091] Each of the battery cells **100** may be provided at one side thereof with terminals **1011**, **1012** electrically connected to the connection tab **1020** and may be formed with a vent **1013**, which is a discharge channel of gas generated therein. The terminals **1011**, **1012** of the battery cell **100** may be a cathode terminal **1011** and an anode terminal **1012**, respectively, and the terminals **1011**, **1012** of adjacent battery cells **100** may be electrically connected in series or in parallel by the connection tab **1020** described below. Although a series connection is described above by way of example, it should be understood that various connection structures may be used, as needed. In addition, it should be understood that the number and arrangement of battery cells are not limited to the structure of FIG. 3 and may be changed, as needed.

[0092] The protection circuit module (PCM) **1030** may have electronic components and protection circuits mounted thereon and may be electrically connected to the connection tabs **1020** described below. The protection circuit module **1030** may include a first protection circuit module **1030a** and a second protection circuit module **1030b** extending from different locations in an arrangement direction of the battery cells **100**, in which the first protection circuit module **1030a** and the second protection circuit module **1030b** may be spaced apart parallel to each other by a constant distance and each may be electrically connected to the connection tabs **1020** adjacent thereto. For example, the first protection circuit module **1030a** extends at one side of upper portions of the plurality of battery cells **100** in the arrangement direction of the battery cells **100** and the second protection circuit module **1030b** extends at the other side of the upper portion of the battery cells **100** in the arrangement direction of the battery cells **100** such that the second protection circuit module **1030b** is spaced apart from the first protection circuit module **1030a**, with the vents **1013** disposed therebetween, while being parallel to the first protection circuit module **1030a**. As such, the two protection circuit modules are spaced apart from each other in the arrangement direction of the battery cells **100**, thereby minimizing the area of the printed circuit board (PCB) constituting the protection circuit module **1030**. An unnecessary PCM area is minimized by dividing the protection circuit module **1030** into two separate protection circuit modules. In addition, the first protection circuit module **1030a** and the second protection circuit module **1030b** may be connected to each other by a conductive connection member **1050**. The connection member **1050** is connected at one side thereof to the first protection circuit module **1030a** and at the other side thereof to the second protection circuit module **1030b**, whereby the two protection circuit modules can be electrically connected.

[0093] The connection may be realized by any one of soldering, resistance welding, laser welding, or projection welding.

[0094] The connection member **1050** may be, for example, an electrical wire. In addition, the connecting member **1050** may include an elastic or flexible material. Through the connection member **1050**, the voltage, temperature, and current of the battery cells **100** can be checked and managed to be normal. That is, information, such as voltage, current, and temperature, received by the first protection circuit module **1030a** from the connection tabs adjacent thereto and information, such as voltage, current, and temperature, received by the second protection circuit module **1030b** from the connection tabs adjacent thereto may be integrated and managed by the protection circuit module through the connection member.

[0095] In addition, upon swelling of the battery cells **100**, impact can be absorbed by elasticity or flexibility of the connection member **1050** to prevent damage to the first and second protection circuit modules **1030a**, **1030b**.

[0096] Further, the shape and structure of the connecting member **1050** are not limited to those shown in FIG. 5.

[0097] As such, the structure of the protection circuit module **1030** divided into the first and second protection circuit modules **1030a**, **1030b** can secure an interior space of the battery module by minimizing the area of the PCB that constitutes the protection circuit module. As a result, it is possible to improve work efficiency by facilitating not only operation of connecting the connection tabs **1020** to the protection circuit module **1030**, but also repair upon detection of failure of the battery module **1000**.

[0098] FIG. 5 illustrates the protection circuit module **1030** applied to the battery module **1000** including a plurality of battery cells **100** illustrated in FIG. 1 to FIG. 4. The protection circuit module **1030** protects and/or manages the battery cells **100** in the battery module **1000**, as described above. For example, the protection circuit module **1030** monitors the state of each of the battery cells **100** to ensure that each of the battery cells **100** is charged or discharged under optimal conditions. Monitoring includes, for example, any behavior that measures and/or manages the voltage, current, temperature, and/or the like of each of the battery cells **100**. Based on these measurements, the protection circuit module **1030** can predict a charge state and/or lifespan of each of the battery cells **100**. For example, the protection circuit module **1030** may control each of the battery cells **100** to prevent occurrence of overcharge, over-discharge and/or overcurrent. Alternatively, for example, the protection circuit module **1030** may act to reduce a deviation between the battery cells **100**.

[0099] The protection circuit module **1030** includes electrical components for measuring the state of the battery cells **100**. The electrical components include, for example, a wire harness, a flexible printed circuit assembly (FPCA), and a flexible die-cut circuit (FDC). The protection circuit module **1030** may include a wire harness as an electrical component when the battery module **1000** includes a relatively small number of battery cells **100**. Alternatively, the protection circuit module **1030** may include an FPCA and/or an FDC as electronic components when the battery module **1000** includes a relatively large number of battery cells **100**.

[0100] The protection circuit module **1030** may include a fuse, which is a safety device to prevent fire from spreading to adjacent circuits in the event of overcurrent and/or electric short in the battery module **1000** and/or each of the battery cells **100**.

[0101] For example, the temperature of one of the battery cells **100** can increase. Herein, a cell having an increased temperature will be referred to as a trigger cell. The fuse identifies (e.g., determines) a trigger cell. The fuse blocks heat propagation between the plurality of battery cells **100**. Specifically, the fuse prevents heat from propagating from the trigger cell to other battery cells by interrupting power between the electrical components.

[0102] The protection circuit module **1030** including such a fuse will be described in more detail. Specifically, the protection circuit module **1030** including a fuse that detects the trigger cell and/or

blocks heat propagation between the battery cells will be described.

[0103] FIG. **6** is a schematic cross-sectional view of a fuse according to one embodiment of the technology described herein.

[0104] A protection circuit module **300** (see FIG. **11**) (for example, including the protection circuit module **1030** described in FIG. **5**) according to one embodiment of the technology described herein includes a substrate **210** with a wire **220** arranged thereon and a fuse formed on the substrate **210**.

[0105] In FIG. **6**, reference numeral **200** denotes a fuse (for example, the fuse illustrated in FIG. **5**) according to one embodiment of the technology described herein. The fuse **200** detects a trigger cell. The fuse **200** also prevents heat from propagating from the trigger cell towards other battery cells.

[0106] To this end, the fuse **200** according to the embodiment of the technology described herein includes the wire **220**, which forms an opening **221** in at least a portion thereof, a solder **230** inserted into the opening **221** to join (e.g., at least partially fill) the opening **221**, and a cover **240** covering at least a portion of the solder **230**.

[0107] The wire **220** is disposed on the substrate **210**. For example, the wire **220** is printed on the substrate **210**.

[0108] The substrate **210** is placed at one side of the plurality of battery cells **100**. The substrate **210** may extend in the arrangement direction of the battery cells **100**. The substrate **210** may be referred to as, for example, a flexible printed circuit assembly (FPCA), a flexible printed circuit board (FPCB), a flexible die-cut circuit (FDC), or the like. The substrate **210** may include, for example, PI, PEN, and the like.

[0109] The wire **220** is disposed on the substrate **210** and is electrically connected to the battery cells **100**, which are electrically connected to the substrate **210**. For example, the wire **220** is electrically connected to the battery cells **100** through busbars. The busbar will be described in more detail with reference to FIG. **10** and FIG. **11**.

[0110] To this end, the wire **220** includes a first metal. The first metal includes a conductive material. The first metal includes, for example, a metal. The first metal includes, for example, copper (Cu), silver (Ag), gold (Au), molybdenum (Mo), cobalt (Co), nickel (Ni), manganese (Mn), platinum (Pt), and/or the like. The wire **220** will be described below with reference to an example in which the wire **220** includes copper (Cu).

[0111] On the other hand, when the first metal melts upon increase in temperature of the battery module **1000**, the voltage supplied to all of the battery cells **100** in the battery module **1000** can be interrupted despite operation of the fuse **200**. Accordingly, the first metal may include, for example, a metal having a relatively high melting point so as not to melt even upon increase in interior temperature of the battery module **1000**. The melting point of the first metal will be described in more detail with reference to FIG. **7**.

[0112] The wire **220** forms the opening **221** in at least a portion thereof. The wire **220** forms an open circuit through the opening **221**.

[0113] The solder **230** is inserted into at least a portion of the opening **221** formed in the wire **220**. The solder **230** is inserted into the opening **221** to join the wire **220** exposed through the opening **221**. For example, the solder **230** includes a first solder **231** inserted into the opening **221**. The first solder **231** may fill the entirety of the opening **221** to allow the wire **220** to be electrically connected to an electric component. Alternatively, the first solder **231** may fill a portion of the opening **221** while allowing the wire **220** to be electrically connected thereto. Alternatively, for example, the solder **230** includes a second solder **232** disposed in a region adjacent to the opening **221**. The second solder **232** is disposed in a region adjacent to an upper portion of the opening **221**, for example, at an upper portion of the wire **220**. The second solder **232** assists in bonding and electrical connection of the wire **220**. The wire **220** allows current to continuously flow through the solder **230** without being interrupted.

[0114] The solder **230** includes a second metal. The second metal includes a conductive material.

The second metal includes, for example, a metal. The second metal may, for example, melt and vaporize to disconnect the wire **220** upon increase in temperature of the battery cell **100**. To this end, the second metal is configured to melt upon increase in temperature of the battery cell **100**. Accordingly, the second metal includes a metal having a lower melting point than the first metal. However, the second metal is configured not to melt upon operation of the battery cells **100** under normal conditions (for example, charging and discharging of the battery cell **100** at 120° C. or less). When the second metal melts in operation of the battery cells **100** under normal conditions, normal operation of the battery cells **100** can be interrupted. Thus, it is desirable that the second metal have a higher melting point than the temperatures of the battery cells **100** under normal operation conditions. For example, the second metal has a melting point of 200° C. or higher. The second metal includes at least one selected from the group consisting of, for example, Al, Mg, Sb, Zn, Pb, Cd, Bi, and Sn.

[0115] The cover **240** covers at least a portion of the solder **230**. For example, the cover **240** is formed over the entire region in which the solder **230** is formed. In top plan view, the cover **240** may have a larger area than the solder **230**. For example, the cover **240** may cover the entire region in which the solder **230** is formed and an upper portion of the wire **220** adjacent to the solder **230**. Alternatively, the cover **240** may also have a smaller area than or the same area as the solder **230** in top plan view.

[0116] The cover **240** provides a migration path of the second metal in the event that the second metal melts, as described below. Accordingly, the cover **240** is configured to maintain the shape thereof and not to melt together when the second metal melts.

[0117] The cover **240** includes a heat resistant material. The cover **240** includes at least one material selected from the group consisting of, for example, polyethylene naphthalate (PEN), polyimide (PI), polyethylene terephthalate (PET), polytetrafluoroethylene (PTFE), polycarbonate (PC), polyethersulphone (PES), and combinations thereof.

[0118] Alternatively, the cover **240** may include an insulating material. The cover **240** may include at least one material selected from the group consisting of, for example, aramid fiber, polybenzimidazole (PBI), polyimide (PI), glass fiber, carbon fiber, grass wool, mineral wool, silica, perlite, urethane foam, ceramic fiber, polytetrafluoroethylene (PTFE), mica, and combinations thereof.

[0119] The cover **240** is formed to a thickness of, for example, 0.015 mm to 0.060 mm. If the thickness of the cover **240** is less than 0.015 mm, the cover **240** may not properly provide the migration path of the second metal. If the thickness of the cover **240** is greater than 0.060 mm, the cover **240** becomes too thick, causing deterioration in capacity of the battery cells **100**. Thus, for example, the cover **240** may be formed to a thickness of 0.015 mm to 0.060 mm. Alternatively, for example, the cover **240** may be formed to a thickness of 0.020 mm to 0.060 mm. Alternatively, for example, the cover **240** may be formed to a thickness of 0.015 mm to 0.055 mm. Alternatively, for example, the cover **240** may be formed to a thickness of 0.020 mm to 0.055 mm. Alternatively, for example, the cover **240** may be formed to a thickness of 0.015 mm to 0.050 mm. Alternatively, for example, the cover **240** may be formed to a thickness of 0.020 mm to 0.050 mm.

[0120] The cover **240** forms a hole **241**. The hole **241** is formed through the cover **240**. The cover **240** may be formed to penetrate from an upper portion of the solder **230** to an upper portion of the cover **240**.

[0121] The cover **240** may include one hole **241** or a plurality of holes **241**. The one hole **241** or at least some of the holes **241** formed in the cover **240** are disposed on the solder **230**. As described above, upon increase in temperature of the battery cell **100**, the solder **230** can melt and vaporize. The holes **241** provide a migration path of the vaporized second metal to allow the vaporized second metal to escape from the solder **230**.

[0122] With this configuration, the fuse **200** according to the embodiment of the technology described herein allows the vaporized second metal to be exposed to the outside. Accordingly, the

fuse **200** can provide a means for interrupting electrical connection to each or all of the battery cells **100** by checking heat propagation due to increase in temperature of the battery cells **100**.

[0123] FIG. 7 is a graph depicting temperature change of a single battery cell and other battery cells adjacent thereto over time upon increase in temperature of the corresponding battery cell.

[0124] FIG. 7 is a graph representing results of an experiment to extract detailed data for each component included in the fuse **200**. The graph of FIG. 7 shows the temperature change of surrounding battery cells over time after the temperature of the trigger cell increases.

[0125] In FIG. 7, the horizontal axis indicates time (sec) and the vertical axis indicates temperature ($^{\circ}$ C.). Here, 0 seconds indicates the time when the trigger cell starts to heat up.

[0126] In FIG. 7, #0 indicates the trigger cell.

[0127] In FIG. 7, #1 indicates a battery cell closest to the trigger cell (hereinafter referred to as “adjacent cell”). In FIG. 7, #2 indicates a battery cell, which is closest to the adjacent cell and is not the trigger cell (hereinafter referred to as “third cell”). In FIG. 7, #3 indicates a battery cell, which is closest to the third cell and is not the adjacent cell (hereinafter referred to as “other cell”).

[0128] As shown in FIG. 7, the trigger cell #0 reaches a peak temperature of 890° C. in the event of heat propagation.

[0129] As shown in FIG. 7, the adjacent cell #1 undergoes a peak temperature of about 710° C. in the event of heat propagation from the trigger cell #0. The third cell #2 also undergoes a peak temperature of about 180° C. in the event of heat propagation from the trigger cell #0. In addition, the other cell #3 has a peak temperature of less than 100° C. even upon heat propagation from the trigger cell #0, thereby exhibiting substantially no temperature change.

[0130] In this case, the wire **220** is required to open with respect to the adjacent cell #1 such that the fuse **200** interrupts electrical connection to the adjacent cell #1. On the other hand, the wire **220** is required not to open with respect to the third cell #2 and/or the other cell #3 such that the fuse **200** does not interrupt electrical connection to the third cell #2 and/or the other cell #3.

[0131] Accordingly, the solder **230** included in the fuse **200** may have a lower melting point than the peak temperature of the adjacent cell #1. Further, the solder **230** may have a higher melting point than the peak temperature of the third cell #2 and the other cell #3. In this case, the peak temperature may have a tolerance of $+20^{\circ}$ C. since the peak temperature can vary depending on situation. Accordingly, for example, the second metal includes a conductive material having a melting point of 200° C. or higher. Table 1 shows melting points ($^{\circ}$ C.) of metals.

TABLE-US-00001 TABLE 1 Metal Melting point ($^{\circ}$ C.) Metal Melting point ($^{\circ}$ C.) Cr $1,890 \pm 10$
Al 660.2 ± 0.1 Pt $1,773.5 \pm 1$ Mg 650 ± 2 Fe $1,539 \pm 3$ Sb 630.5 ± 0.1 Co $1,495 \pm 1$ Zn 419.46 Ni
 $1,455 \pm 1$ Pb 327.4 ± 0.1 Mn $1,245 \pm 10$ Cd 320.9 ± 0.1 Cu $1,083 \pm 0.1$ Bi 271.3 ± 0.1 Au $1,063$
Sn 231.9 ± 1 Ag 960.5

[0132] As can be seen from Table 1, metals having melting points that are lower than the peak temperatures of the adjacent cell #1 and higher than the peak temperature of the third cell #2 and the other cell #3 include, for example, Al, Mg, Sb, Zn, Pb, Cd, Bi, Sn, and the like. Thus, the second metal includes at least one material selected from the group consisting of, for example, Al, Mg, Sb, Zn, Pb, Cd, Bi, Sn, and alloys thereof.

[0133] Here, when the wire **220** includes Cu, the second metal may be, for example, Pb. This is because Cu and Pb have good bonding reliability and Pb can be prepared by simply applying solder pastes in an SMT process.

[0134] As shown in FIG. 7, it can be seen that the third cell #2 reaches the peak temperature after 13.5 seconds (sec) have elapsed. Based on such experimental data, each of the components included in the fuse **200** will be described in detail with reference to FIG. 8 and FIG. 9.

[0135] FIG. 8 is a schematic cross-sectional view of a fuse according to one embodiment of the technology described herein.

[0136] FIG. 8 shows the width of the opening **221** formed in the wire **220**. As described with reference to FIG. 6, the wire **220** may be open through the opening **221**. For convenience of

description, a portion of the wire **220** placed at one side of the opening **221** will be referred to as a first wire **220a** and a portion of the wire **220** placed at the other side of the opening **221** will be referred to as a second wire **220b**. Here, the width x of the opening **221** indicates the shortest distance between the first wire **220a** and the second wire **22b**.

[0137] Referring to FIG. 7, it can be seen that the third cell #2 does not reach the peak temperature thereof until 13.5 seconds after the trigger cell starts to generate heat. Thus, the fuse **200** according to the embodiment of the technology described herein can protect most of the battery cells **100** in the battery module **1000** by interrupting electrical connection to the trigger cell before the third cell #2 reaches the peak temperature thereof.

[0138] To this end, the wire **220** may be formed to a width of 4 mm or less. When the width x of the wire **200** exceeds 4 mm, the amount of solder **230** inserted into the opening **221** can become excessive. In this case, it can take a long period of time for the solder **230** to melt and form an open circuit. Alternatively, the wire **220** may be formed to a width of, for example, 0.1 mm or more.

Alternatively, the wire **220** may be formed to a width of, for example, 0.2 mm or more.

Alternatively, the wire **220** may be formed to a width of, for example, 0.3 mm or more.

Alternatively, the wire **220** may be formed to a width of, for example, 0.4 mm or more.

Alternatively, the wire **220** may be formed to a width of, for example, 0.5 mm or more.

Alternatively, the wire **220** may be formed to a width of, for example, 0.6 mm or more.

Alternatively, the wire **220** may be formed to a width of, for example, 0.7 mm or more.

Alternatively, the wire **220** may be formed to a width of, for example, 0.8 mm or more.

Alternatively, the wire **220** may be formed to a width of, for example, 0.9 mm or more.

Alternatively, the wire **220** may be formed to a width of, for example, 1.0 mm or more. When the width of the wire **220** is less than 0.1 mm, the distance between the first wire **220a** and the second wire **220b** can become too close for the wire **220** to form an open circuit even when the solder **230** melts. Thus, the wire **220** may be formed to a width of, for example, 0.1 mm to 4 mm.

Alternatively, the wire **220** may be formed to a width of 1.0 mm to 4 mm.

[0139] With this structure, the fuse **200** according to the embodiment of the technology described herein can promptly detect and prevent heat propagation from the trigger cell toward the battery cells within the battery module **1000**.

[0140] Furthermore, the fuse **200** according to the embodiment of the technology described herein can prevent heat propagation from the battery cell **100**, which is not provided with a temperature sensor, through the solder **230** and the cover **240**, even upon increase in temperature of the battery cell **100**.

[0141] FIG. 9 is a schematic top plan view of the fuse according to the embodiment of the technology described herein.

[0142] FIG. 9 illustrates an area ratio of the fuse. Here, the area ratio of the fuse is based on the area of the fuse **200** in top plan view. The area ratio of the fuse is calculated according to Equation 1:

Area ratio=Area of hole/area of solder.

[0143] Here, the “area of hole” indicates the sum of areas of one or more holes **241** placed on the solder **230**. For example, if the cover **240** includes three holes, two of which are placed on the solder **230** and the other one is not placed on the solder **230**, the “hole” in the “area of hole” refers to the two holes placed on the solder **230**.

[0144] For example, the holes **241** may be formed in a rectangular shape, as shown in FIG. 9. Here, each of the holes **241** may have a length $b1$ at one side and a length $b2$ at another side. In this case, the area of one hole **241** is calculated by $b1 \times b2$. FIG. 9 illustrates an example where the hole **241** includes a first hole **2411** and a second hole **2412**, which have the same area and are spaced apart from each other and placed on the solder **230**. In this case, the “area of hole” may be calculated by $b1 \times b2 \times 2$. However, it should be understood that this is provided only by way of example and the

holes **241** may be formed not only in a rectangular shape, but also in a polygonal, circular, elliptical, or irregular shape. Furthermore, the cover **240** may be formed with a single hole **241** or at least three holes **241**. The area of hole may be calculated differently depending on the shape and/or the number of holes.

[0145] The “area of solder” indicates an area of an upper surface of the solder **230**. For example, when the solder **230** includes the second solder **230** as shown in FIG. **6**, the area of solder may be the area of the upper surface of the second solder **230**.

[0146] For example, the upper surface of the solder **230** may have a length $a1$ at one side and a length $a2$ at another side. In this case, the area of the solder **230** is calculated by $a1 \times a2$. However, it should be understood that this is provided only by way of example and the area of solder may be calculated differently depending on the shape of the solder.

[0147] The fuse **200** according to the embodiment of the technology described herein has an area ratio of, for example, 0.4 to 0.65. If the area ratio of the fuse **200** is less than 0.4, it can take a long period of time for vaporization of the solder **230**. In this case, the fuse **200** may not provide an open circuit within a suitable period of time. A suitable period of time is, for example, 13.5 seconds, as described with reference to FIG. **7** and FIG. **8**. If the area ratio of the fuse exceeds 0.65, the cover **240** can suffer from cracking. Accordingly, the fuse may be formed to have an area ratio of 0.4 to 0.65.

[0148] With this structure, the fuse **200** according to the embodiment of the technology described herein can promptly detect and prevent heat propagation from the trigger cell toward the battery cells within the battery module **1000**.

[0149] Furthermore, the fuse **200** according to the embodiment of the technology described herein can prevent heat propagation from the battery cell **100**, which is not provided with a temperature sensor, through the solder **230** and the cover **240**, even upon increase in temperature of the battery cell **100**.

[0150] The fuse **200** according to the embodiment of the technology described herein has been described with reference to FIG. **6** to FIG. **9**. Hereinafter, the protection circuit module **300** including such a fuse **200** and a battery module **1000** including the protection circuit module **300** will be described.

[0151] FIG. **10** is a top plan view of a battery module according to one embodiment of the technology described herein.

[0152] FIG. **11** is an enlarged view of Region A in FIG. **10**.

[0153] As illustrated in FIG. **5**, the battery module **1000** according to the embodiment of the technology described herein includes the plurality of battery cells **100** and the housing **1061** to **1065** that receives the plurality of battery cells **100** therein. The battery module **1000** further includes a protection circuit module **300** at one side of the plurality of battery cells **100**. FIG. **10** is a top plan view of the battery module **1000** to which the protection circuit module **300** is applied.

[0154] Referring to FIG. **11**, the battery cells **100** are electrically connected to the outside through a cap plate **250**. The protection circuit module **300** includes a connection plate **260**. The connection plate **260** is electrically connected to the cap plate **250**. The connection plate **260** may be connected to the cap plate **250** by welding.

[0155] In this case, the fuse **200** is placed within a certain distance (Y) from the connection plate **260**. If the fuse **200** is placed too far from the connection plate **260**, the temperature of the solder **230** may not increase significantly despite increase in temperature of the trigger cell. In this case, the solder **230** may not melt despite heat dissipation from the trigger cell. Here, the position of the fuse **200** is determined relative to the position of the solder **230**. The position of the solder **230** is set with reference to a center of the solder **230** or a center of the width of the opening.

[0156] In consideration of such a feature, for example, the fuse **200** may be placed at a distance of 40 mm or less from the connection plate **260**. Here, a reference position of the connection plate **260** for the distance of 40 mm is a point on the connection plate **260**, which is placed at the shortest

distance from the fuse **200**, for example, on a centerline parallel to the longitudinal direction of the connection plate **260**. If the fuse **200** is placed at a distance of greater than 40 mm from the connection plate **260**, the temperature of the solder **230** may not increase above the melting point thereof within 13.5 seconds even upon heat propagation from the trigger cell. In this case, the fuse **200** can fail to form an open circuit within 13.5 seconds. For example, the fuse **200** is placed at a distance of 10 mm to 40 mm from the connection plate **260**.

[0157] With this structure, the fuse **200** according to the embodiment of the technology described herein can promptly detect and prevent heat propagation from the trigger cell toward the cells within the battery module **1000**.

[0158] Next, comparative examples and examples of the technology described herein will be described with reference to Tables 2 and 3.

TABLE-US-00002

TABLE 2	Type X	Circuit	Circuit open time (sec)	Example 1	1 mm	PASS	12.20
Example 2	2 mm	PASS	12.30	Example 3	3 mm	PASS	12.70
Example 4	4 mm	PASS	13.10	Comparative Example 1	5 mm	NG	13.50
Comparative Example 2	6 mm	NG	14.20	Comparative Example 3	7 mm	NG	15.50

[0159] In Table 2, Examples 1 to 4 and Comparative Examples 1 to 3 provide fuses all having the same conditions except for the width x. Specifically, in Examples 1 to 4 and Comparative Examples 1 to 3, the first metal is composed of Cu and the second metal is composed of Pb.

[0160] In Table 2, “x” indicates the width of the fuse. “Circuit” indicates the wire **220** of the third cell. “Circuit open time (sec)” indicates a time it takes for the wire **220** of the trigger cell to open.

[0161] Examples 1 to 4 provide fuses **200** according to the embodiment of the technology described herein, in which each wire **220** has a width x of 1 mm to 4 mm. As can be seen from Table 2, in Examples 1 to 4, the circuit opens within 13.5 seconds (not including 13.5 seconds). In this case, as described in FIG. 7 and FIG. 8, the fuses **200** can interrupt electrical connection of the trigger cell before the third cell reaches the peak temperature thereof.

[0162] Comparative Examples 1 to 3 provide fuses in which each wire **220** has a width x of 5 mm to 7 mm. As can be seen from Table 2, in Comparative Examples 1 to 3, the circuit opens after 13.5 seconds (including 13.5 seconds). In this case, as described in FIG. 7 and FIG. 8, the fuses interrupt electrical connection of the trigger cell after the adjacent cell has reached the peak temperature thereof. That is, the fuses of Comparative Examples 1 to 3 perform late voltage interruption with respect to the trigger cell.

TABLE-US-00003

TABLE 3	Fusing Time	Presence of Type	Area ratio (sec)	cracks	Fusing Spec
Comparative	0.25	42.0	Absent	NG	Example 4
Comparative	0.30	31.0	Absent	NG	Example 5
Comparative	0.35	21.0	Absent	NG	Example 6
Example 5	0.40	12.0	Absent	Okay	Example 6
Example 6	0.45	8.5	Absent	Okay	Example 7
Example 7	0.50	8.1	Absent	Okay	Example 8
Example 8	0.55	7.5	Absent	Okay	Example 9
Example 9	0.60	7.3	Absent	Okay	Example 10
Example 10	0.65	7.1	Absent	Okay	Comparative
Comparative	0.70	6.8	Present	Okay	Example 7
Comparative	0.75	6.8	Present	Okay	Example 8

[0163] In Table 3, all of Examples 5 to 10 and Comparative Examples 4 to 8 have the same conditions except for the area ratio. Specifically, in Examples 5 to 10 and Comparative Examples 4 to 8, the first metal is composed of Cu and the second metal is composed of Pb.

[0164] In Table 3, “area ratio” indicates a value calculated by Equation 1 as described in FIG. 9. In Table 3, “fusing time (sec)” indicates a time it takes for the fuse **200** of the trigger cell to form an open circuit. In Table 3, “presence of cracks” indicates whether or not cracking has occurred in the cover **240**. “Fusing Spec” indicates whether the fusing time (sec) falls within 13.5 seconds.

[0165] Examples 5 to 10 provide fuses according to embodiments of the technology described herein, in which the area ratio ranges from 0.4 and 0.65. As can be seen from Table 3, in Examples 5 to 10, the fusing time (sec) is within 13.5 seconds. Furthermore, in Examples 5 to 10, cracking does not occur.

[0166] Comparative Examples 4 to 6 provide fuses having an area ratio of less than 0.4. As can be seen from Table 3, in Comparative Examples 4 to 6, the fusing time exceeds 13.5 seconds (sec).

That is, the fuses of Comparative Examples 4 to 6 interrupt electrical connection of the trigger cell after the adjacent cell has reached the peak temperature thereof. Therefore, the fuses of Comparative Examples 4 to 6 perform late voltage interruption with respect to the trigger cell. [0167] Comparative Examples 7 and 8 provide fuses having an area ratio of greater than 0.65. As can be seen in Table 3, in Comparative Examples 7 and 8, cracking occurs. That is, Comparative Examples 7 and 8 fail to provide the cover **240** and thus do not provide the fuse **200** according to the embodiment of the technology described herein.

[0168] Although the technology described herein has been described with reference to some embodiments and drawings illustrating aspects thereof, the technology described herein is not limited thereto. Various modifications and variations can be made by a person skilled in the art to which the technology described herein belongs within the scope of the technical spirit of the technology described herein and the claims and equivalents thereto.

Claims

1. A protection circuit module comprising: a substrate; a wire disposed on the substrate to form an opening in at least a portion of the wire and the wire comprising a first metal; a solder comprising a second metal having a melting point lower than a melting point of the first metal and the second metal being inserted into at least a portion of the opening to join ends of the wire separated by the opening; and a cover forming at least one hole and covering at least a portion of the solder.
2. The protection circuit module according to claim 1, wherein the at least one hole is placed on the solder and provides a migration path for vaporized second metal upon vaporization of the second metal.
3. The protection circuit module according to claim 1, wherein the opening has a width of 1 mm to 4 mm.
4. The protection circuit module according to claim 1, wherein the melting point of the second metal is 200° C. or higher.
5. The protection circuit module according to claim 1, wherein the second metal comprises at least one material selected from the group consisting of Al, Mg, Sb, Zn, Pb, Cd, Bi, and Sn.
6. The protection circuit module according to claim 1, wherein the cover comprises at least one material selected from the group consisting of polyethylene naphthalate (PEN), polyimide (PI), polyethylene terephthalate (PET), polytetrafluoroethylene (PTFE), polycarbonate (PC), and polyethersulphone (PES).
7. The protection circuit module according to claim 1, wherein the cover has a thickness of 0.015 mm to 0.060 mm.
8. The protection circuit module according to claim 1, wherein an area ratio of the at least one hole to the solder ranges from 0.4 to 0.65 in top plan view.
9. The protection circuit module according to claim 1, further comprising: a connection plate electrically connected to a battery cell, wherein the opening is formed within 40 mm from the connection plate.
10. A battery module comprising: a plurality of battery cells; a housing for receiving the plurality of battery cells; and a protection circuit module disposed at one side of the plurality of battery cells, the protection circuit module comprising: a substrate; a wire disposed on the substrate to form an opening in at least a portion of the wire and the wire comprising a first metal; a solder comprising a second metal having a melting point lower than a melting point of the first metal and the second metal being inserted into at least a portion of the opening to join ends of the wire separated by the opening; and a cover forming at least one hole and covering at least a portion of the solder.
11. The battery module according to claim 10, wherein the at least one hole is placed on the solder and provides a migration path for vaporized second metal upon vaporization of the second metal.
12. The battery module according to claim 10, wherein the opening has a width of 1 mm to 4 mm.

13. The battery module according to claim 10, wherein the melting point of the second metal is 200° C. or higher.

14. The battery module according to 10, wherein the second metal comprises at least one material selected from the group consisting of Al, Mg, Sb, Zn, Pb, Cd, Bi, and Sn.

15. The battery module according to claim 10, wherein the cover comprises at least one material selected from the group consisting of polyethylene naphthalate (PEN), polyimide (PI), polyethylene terephthalate (PET), polytetrafluoroethylene (PTFE), polycarbonate (PC), and polyethersulphone (PES).

16. The battery module according to claim 10, wherein the cover has a thickness of 0.015 mm to 0.060 mm.

17. The battery module according to claim 10, wherein an area ratio of the at least one hole to the solder ranges from 0.4 to 0.65 in top plan view.

18. The battery module according to claim 10, wherein the protection circuit module further comprises a connection plate electrically connected to the plurality of battery cells and the opening is formed within 40 mm from the connection plate.
